

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of semester 2

Year: 2019

Subject: 394172 Mathematics II

Section: 15-18

Date: 21 January 2020

Time: 08.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. This test is divided into 9 questions. They will be 9 pages (including this page) and they are worth 90 points.

1. Answer the questions on the test paper.
2. Book, documents and lecture notes are not allowed.
3. You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
4. Before leaving, make sure you do not bring the test outside.
5. Do not use electronic communication device.
6. Calculations are NOT allowed in the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

No.	1.	2.	3.	4.	5.	6.	7.	8.
score								

Question 1 Let p, q, r, s and t be statements.

1.1 Given p is true, $r \vee \sim p$ is true, and $p \rightarrow (q \rightarrow r)$ is true.

Determine the truth value of $[p \rightarrow (q \rightarrow \sim r)] \leftrightarrow \sim (q \wedge r)$. (5 points)

1.2 The following truth values of $p \rightarrow (q \wedge r)$ is false and $p \leftrightarrow (s \vee t)$ is true.
Find the truth value of $(q \wedge s) \rightarrow (p \wedge q)$. (5 points)

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Question 2 Show whether the following statements are logically equivalent or not.

2.1 $(s \rightarrow r) \wedge (q \rightarrow r)$ and $(s \vee q) \rightarrow r$. (5 points)

2.2 Show whether the statement $[(\sim p \wedge \sim q) \rightarrow q] \rightarrow (\sim q \rightarrow p)$ is tautology or not.
(5 points)

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Question 3 3.1 Given $U = \{x \in \mathbb{R} \mid 0 < x < 1\}$. Find the negation of the quantified statement $\exists x \forall y [x^2 - y^2 < y - x]$ and determine whether the statement or its negation is true. (5 points)

3.2 Given the argument is in the form

- Hypothesis
1. $p \rightarrow q$
 2. $q \rightarrow r$
 3. $p \vee s$
 4. $\sim s \vee r$

Conclusion: r .

Show whether this argument is valid or not. (5 points)

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Question 4 4.1 Convert the following numbers to a number in the base-10. (6 points)

a) $10101.01_2 =$

b) $AE7_{16} =$

4.2 Convert $(A3.F)_{16}$ to a number in the base-8. (4 points)

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Question 5. Convert the following expressions to a number with the given base.

5.1) $104_{12} - 312_5$ (base-7) (4 points)

5.2) $(2134_8 + 756_8) \div 43_8$ (base-8) (6 points)

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Question 7 7.1 Find the values of $a + b + c$ that satisfies the expression
 $(39.A2)_{16} = (7a.604)_8 = (b21.300c)_4$. (5 points)

7.2 Solve the absolute value inequality $x^2 + 10|x| > 39$. (5 points)

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Question 8 8.1 Find all solutions of the absolute value inequality $\frac{3|x-2|}{2-x} < 1-x$.
(5 points)

8.2 Find all solutions of $\frac{|x|-x^2}{x-|x-1|} \geq 0$. (5 points)

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Question 6 6.1 To simplify the number $[(3013)_4 - (157)_8] \times (49)_{16}$ and convert this answer in base-16. (5 points)

6.2 Convert the number $(0.45)_8$ to the number in base -4. (5 points)

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